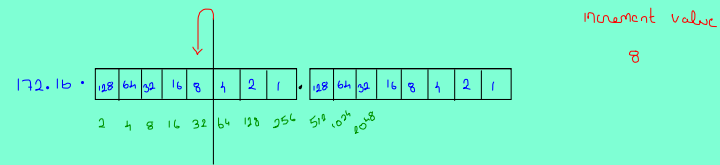


172.16.0.0/16 Req 30 Dept



172.16.0.0/21

old Subnet mask

255.255.0.0

172.16.8.0/21

new Subnet mask

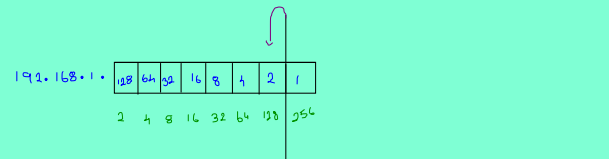
172.16.16.0/21

255.255.248.0

⋮

172.16.248.0/21

192.168.1.0/24 Req 120 Dept



192.168.1.0/31

255.255.255.254

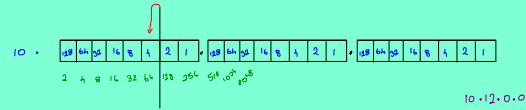
192.168.1.2/31

192.168.1.4/31

⋮

192.168.1.254

10.0.0.0/8 Req. 50 Dept



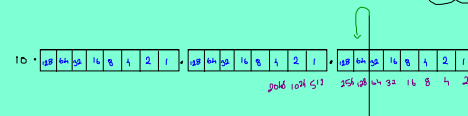
	10.0.0.0/14	10.4.0.0/14	10.8.0.0/14
First IP	10.0.0.1	10.4.0.1	10.8.0.1
Last IP	10.3.255.254	10.7.255.254	10.11.255.254
Broadcast IP	10.3.255.255	10.7.255.255	10.11.255.255

Host based Subnetting :-

Step 2: Write the binary combination for Host bit from Right to Left

10.0.0.0/8 Req. 60 IP

Increment = 64



10.0.0.0/26

10.0.0.64

10.0.0.1

10.0.0.2

...

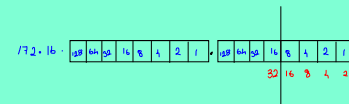
10.0.0.62

10.0.0.63

255.255.255.192

172.16.0.0/16 Req. 10 IP

16



172.16.0.0/28

172.16.0.16

172.16.0.1 -> First

172.16.0.2

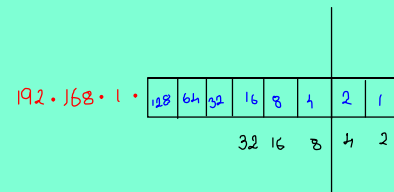
...

172.16.0.14 -> Last IP

172.16.0.15 -> Broadcast

255.255.255.240

192.168.1.0/24 Req 2 IP



192.168.1.0/30

192.168.1.4

192.168.1.1 → First IP

192.168.1.2 → Last IP

255.255.255.252

192.168.1.3 → Broadcast

10.0.0.0/8 Req 1000 IP

172.16.0.0/16 Req 110 Dept

192.168.1.0/24 Req 100 IP

30.1.0.1/8 Req 30 IP

172.100.0.0/16 Req 2000 IP

Step₁ :- Separate the Host Portion with Proper Place value

Step₂ :- write the binary combination For host bit
From Left to Right

Step₃ :- move the Line Based on your Req

Step₄ :- write the Binary combination of new network bit